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Research Positions

- from 09.2027  **Marie Skłodowska-Curie Fellow**, Czech Academy of Sciences, Prague - Czechia.
Host: Prof. Dr. Vladimír Karas.
- since 10.2024  **Alexander von Humboldt/MPA Fellow**, Max Planck Institute for Astrophysics, Garching - Germany. Host: Prof. Dr. Selma de Mink.
- 03.2023 - 09.2024  **Postdoctoral Scientific Staff**, University of Hamburg, Hamburg - Germany.
Adviser: Prof. Dr. Stephan Rosswog.
- 10.2019 - 02.2023  **Postdoctoral Researcher**, Charles University, Prague - Czechia.
Adviser: Assoc. Prof. Dr. Ondřej Pejcha.

Education

- Sep 2019  **Ph.D. in Astrophysics**, *summa cum laude*, Pontificia Universidad Católica de Chile.
Thesis: *"The Role of Interacting Stellar Winds Feeding Sagittarius A*"*.
Advisers: Prof. Dr. Jorge Cuadra & Dr. Marc Schartmann.
- Mar 2015  **Master in Astrophysics**, *summa cum laude*, Pontificia Universidad Católica de Chile.
Thesis: *"Thin-shell instabilities and clump formation due to colliding stellar winds in the Galactic Centre"*.
Adviser: Prof. Dr. Jorge Cuadra.
- Dec 2012  **Bachelor in Astronomy**, *summa cum laude*, Universidad de Concepción, Chile.
Ranked 1 out of 35 enrolled at the Astronomy program in 2009.

Honors and Awards

- Feb 2026  **Marie Skłodowska-Curie Fellowship**, European Commission, European Union.
- Jul 2024  **Heraeus Fellowship**, Wilhelm and Else Heraeus Foundation, Germany.
- Sep 2023  **Quantum Universe Postdoctoral Grant**, University of Hamburg, Germany.
- Mar 2023  **Humboldt Research Fellowship**, Alexander von Humboldt Foundation, Germany.
- 2015-2019  **Doctoral Fellowship**, National Agency for Research and Development (ANID), Chile.
- Mar 2015  **Doctoral Excellence Award**, Pontificia Universidad Católica de Chile.
- 2013-2014  **Master Fellowship**, National Agency for Research and Development (ANID), Chile.
- 2009-2012  **Undergraduate Scholarship**, Center for Excellence in Astrophysics and Associated Technologies (CATA).

Refereed Publications

Summary: 18 published (+2 submitted, +2 in prep.), 9 first-author papers (+2 in prep.).

NASA ADS (July 2026): 228 citations. Papers with students are shown in bold italic.

- 1 Abd El Dayem, K., Abuter, R., Aimar, N., P., A., Berdeu, A., Berger, J.-P., Bourdarot, G., Brandner, W., Burkert, A., **Calderón, D.**, Correia, C., Cuadra, J., Davies, R., Defrère, D., Delit, L., Drescher, A., Eisenhauer, F., Esteras Otal, L., Fabricius, M., ... Zins, G. (2026). Discovery of a star sensitive to the spin of Sgr A*. *Submitted to Nature*.  <https://doi.org/10.21203/rs.3.rs-8619199/v1>
- 2 **Calderón, D.**, Pejcha, O., Metzger, B. D., Duffell, P. C., & Rosswog, S. (2026). Quantifying the Impact of Relativistic Precession on Tidal Disruption Event Light Curves. *Astronomische Nachrichten*, 347(5), Article e70092, e70092.  <https://doi.org/10.1002/asna.70092>

- 3 Gillessen, S., Eisenhauer, F., Cuadra, J., Genzel, R., **Calderón, D.**, Joharle, S., Piran, T., Ribeiro, D. C., Russell, C. M. P., Sadun Bordoni, M., Burkert, A., Bourdarot, G., Drescher, A., Mang, F., Ott, T., Agapito, G., Agudo Berbel, A., Baruffolo, A., Bonaglia, M., ... Xompero, M. (2026). The gas streamer G1-2-3 in the Galactic Center. *A&A*, 707, A79. <https://doi.org/10.1051/0004-6361/202555808>
- 4 **Kirilov, A.**, **Calderón, D.**, Pejcha, O., & Duffell, P. C. (2025). Two-dimensional Radiation-hydrodynamic Simulations of Luminous Red Novae. *ApJL*, 994(2), L41. <https://doi.org/10.3847/2041-8213/ae1ae7>
- 5 **Shaqil, W.**, **Calderón, D.**, Rosswog, S., Cuadra, J., Ciurlo, A., Morris, M. R., Campbell, R. D., & Ghez, A. M. (2025). Tidal phenomena in the Galactic Center: The Curious Case of X7. *A&A*, 700, A134. <https://doi.org/10.1051/0004-6361/202555191>
- 6 Villanueva, V., Blaña, M., Bolatto, A. D., Rubio, M., Tarantino, E., Herrera-Camus, R., Burkert, A., Vaz, D., Read, J. I., Galaz, G., Muñoz, C., **Calderón, D.**, Behrendt, M., Carballo-Bello, J. A., Gray, E., & Fellhauer, M. (2025). The CHIMERA Survey: The first CO detection in Leo T, the lowest mass known galaxy still hosting cold molecular gas. *A&A*, 699, L11. <https://doi.org/10.1051/0004-6361/202555635>
- 7 **Calderón, D.**, Cuadra, J., Russell, C. M. P., Burkert, A., Rosswog, S., & Balakrishnan, M. (2025). The formation and stability of a cold disc made out of stellar winds in the Galactic centre. *A&A*, 693, A180. <https://doi.org/10.1051/0004-6361/202452800>
- 8 Blaña, M., Burkert, A., Fellhauer, M., **Calderón, D.**, Behrendt, M., & Schartmann, M. (2024). The Milky Way satellite galaxy Leo T: A perturbed cored dwarf. *A&A*, 692, A183. <https://doi.org/10.1051/0004-6361/202451649>
- 9 Balakrishnan, M., Corrales, L., Markoff, S., Nowak, M., Haggard, D., Wang, Q. D., Neilsen, J., Russell, C. M. P., **Calderón, D.**, Cuadra, J., & Baganoff, F. (2024). Multistructured Accretion Flow of Sgr A*. I. Examination of a Radiatively Inefficient Accretion Flow Model. *ApJ*, 974(1), 98. <https://doi.org/10.3847/1538-4357/ad6c08>
- 10 Balakrishnan, M., Russell, C. M. P., Corrales, L., **Calderón, D.**, Cuadra, J., Haggard, D., Markoff, S., Neilsen, J., Nowak, M., Wang, Q. D., & Baganoff, F. (2024). Multistructured Accretion Flow of Sgr A*. II. Signatures of a Cool Accretion Disk in Hydrodynamic Simulations of Stellar Winds. *ApJ*, 974(1), 99. <https://doi.org/10.3847/1538-4357/ad6866>
- 11 **Calderón, D.**, Pejcha, O., Metzger, B. D., & Duffell, P. C. (2024). The effect of relativistic precession on light curves of tidal disruption events. *MNRAS*, 528(2), 2568–2587. <https://doi.org/10.1093/mnras/stae194>
- 12 Pejcha, O., **Calderón, D.**, & Kurfürst, P. (2022). Supernovae in colliding-wind binaries: observational signatures in the first year. *MNRAS*, 510(3), 3276–3290. <https://doi.org/10.1093/mnras/stab3729>
- 13 **Calderón, D.**, Pejcha, O., & Duffell, P. C. (2021). Moving-mesh radiation-hydrodynamic simulations of wind-reprocessed transients. *MNRAS*, 507(1), 1092–1105. <https://doi.org/10.1093/mnras/stab2219>
- 14 Gómez, P. L., & Calderón, D. (2020). The Dynamics of the Wide-angle Tailed (WAT) Galaxy Cluster A562. *AJ*, 160(4), 152. <https://doi.org/10.3847/1538-3881/aba831>
- 15 **Calderón, D.**, Cuadra, J., Schartmann, M., Burkert, A., Prieto, J., & Russell, C. M. P. (2020). Three-dimensional simulations of clump formation in stellar wind collisions. *MNRAS*, 493(1), 447–467. <https://doi.org/10.1093/mnras/staa090>
- 16 **Calderón, D.**, Cuadra, J., Schartmann, M., Burkert, A., & Russell, C. M. P. (2020). Stellar Winds Pump the Heart of the Milky Way. *ApJL*, 888(1), L2. <https://doi.org/10.3847/2041-8213/ab5e81>
- 17 **Calderón, D.**, Cuadra, J., Schartmann, M., Burkert, A., Plewa, P., Eisenhauer, F., & Habibi, M. (2018). The Galactic Centre source G2 was unlikely born in any of the known massive binaries. *MNRAS*, 478(3), 3494–3505. <https://doi.org/10.1093/mnras/sty1330>

- 18 **Calderón, D.**, Bauer, F. E., Veilleux, S., Graciá-Carpio, J., Sturm, E., Lira, P., Schulze, S., & Kim, S. (2016). Searching for molecular outflows in hyperluminous infrared galaxies. *MNRAS*, 460(3), 3052–3062. <https://doi.org/10.1093/mnras/stw1210>
- 19 **Calderón, D.**, Ballone, A., Cuadra, J., Schartmann, M., Burkert, A., & Gillessen, S. (2016). Clump formation through colliding stellar winds in the Galactic Centre. *MNRAS*, 455(4), 4388–4398. <https://doi.org/10.1093/mnras/stv2644>

[ADS library link](#)

Service & Skills

Coding	■ C, FORTRAN, Python, SHELL scripts.
Languages	■ Czech (B1), English (C1), German (A2), Spanish (mother tongue).
Professional	■ SOC member at XIX Chilean Astronomical Society Annual Meeting (Nov 2024). ■ SOC member at 2nd Workshop of Astronomy Students (March 2017).
Referee	■ Astronomy & Astrophysics (1), Astrophysical Journal (3), Czech Science Agency (3), Israel Science Foundation.
Representation	■ Postdoctoral Representative 2025-2026 at Max Planck Institute for Astrophysics.
Supervision	■ Timea Unold - Master student at Technical University of Munich - since Oct 2025. ■ Anthony Kirilov - Ph.D. student at Charles University - expected Aug 2026. ■ Wasif Shaqil - Master student at University of Hamburg - Apr 2025. Next position Ph.D. student at Liverpool John Moores University, United Kingdom.
Thesis Committee	■ Jakub Hadač - Bachelor in Physics at Charles University - June 2026. ■ Michal Jireš - Bachelor in Physics at Charles University - Feb 2024. ■ Yelyzaveta Pulnova - Master in Physics at Charles University - Sep 2021.

Teaching Experience

2024-2025	■ Teaching assistant at Ludwig Maximilian University of Munich in graduate course: "Astrophysics Seminar" (Fall 2024, Spring 2025, Fall 2025).
2023-2024	■ Teaching assistant at University of Hamburg in undergraduate course: "Introduction to Astronomy" (Spring 2023); and graduate course: "Astrophysical Hydrodynamics" (Fall 2023).
2019 - 2023	■ Co-organiser of Science Coffee (also as a class), Charles University.
Oct 2021	■ Special lecture: "Introduction to Radiation Hydrodynamics", Charles University.
2017	■ Science Coffee (in charge once per week), Pontificia Universidad Católica de Chile.
2015-2017	■ Teaching Assistant at Pontificia Universidad Católica de Chile in undergraduate courses: "Extragalactic Astrophysics" (Fall 2017), "Introduction to Physics" (Spring 2015), "Introduction to Astronomy" (Fall 2015); and graduate courses: "Galactic Centre Astrophysics" (Spring 2016), "Radiative Processes in Astrophysics" (Fall 2016).
2011-2012	■ Teaching Assistant at Universidad de Concepción in undergraduate courses: "General Astrophysics" (Fall 2012), and "Electromagnetic Theory" (Spring 2011).

Funding and Grants

Feb 2026	■ Marie Skłodowska-Curie Fellowship (~200 k€), European Commission, Europe.
2025-2026	■ Cycle 4 JWST Theory grant #8655 (PI: C. Russell) as co-Investigator, USA.
2025-2028	■ FONDECYT Regular 2025 (PI: J. Cuadra), as co-Investigator, ANID ¹ , Chile.
Jul 2024	■ Heraeus Fellowship (~5 k€), Wilhelm and Else Heraeus Foundation, Germany.

¹ANID: National Agency for Research and Development

Funding and Grants (continued)

Sep 2023	■	Quantum Universe Postdoctoral Grant (10 k€), University of Hamburg, Germany.
Mar 2023	■	Humboldt Fellowship (~100 k€), Alexander von Humboldt Foundation, Germany.
Jan 2023	■	2,210,400 CPU hours awarded , IT4Innovations Supercomputing Center, Czechia.
2022-2023	■	Cycle 24 Chandra Theory grant #24200412 (PI: C. Russell) as co-Investigator, USA.
Mar 2022	■	936,000 CPU hours , IT4Innovations National Supercomputing Center, Czechia.
2021-2022	■	Cycle 23 Chandra Theory grant #23610258 (PI: C. Russell) as co-Investigator, USA.
2015-2019	■	Doctoral Fellowship (~45 k€), ANID, Chile. - Plus internship abroad for 10 months and travel budget for 2 years (total: ~20 k€).
Mar 2015	■	Doctoral Excellence Award (~10 k€), Pontificia Universidad Católica de Chile.
2014,2016,2018	■	Adelina Gutiérrez Travel Grant , Chilean Astronomical Society (SOCHIAS).
2013-2014	■	Master Fellowship (~15 k€), ANID, Chile.
2009-2012	■	Undergraduate Scholarship (~8 k€), Center for Excellence in Astrophysics and Associated Technologies (CATA).

Internships, Extended Workshops, and Summer Schools

Aug 2026	■	MIAPbP program: <i>Bridging Gaps in High-Mass Star Research: Physical Parameters, Formation and Evolution</i> , Garching – Germany.
Jul 2024	■	Special Meeting: 73rd Lindau Nobel Laureate Meeting, Lindau – Germany.
Jul 2023	■	School: Kavli Summer Program in Astrophysics, Garching – Germany.
Jul 2022	■	Workshop: <i>"Plasma Under Strong Gravity"</i> , Aspen Center for Physics, Aspen – USA.
Feb - Nov 2018	■	Visiting Ph.D. Student , Max Planck Institute for Extraterrestrial Physics - Germany. Hosts: Prof. Dr. Andreas Burkert & Dr. Marc Schartmann.
Aug 2017	■	School: <i>"Astrophysical Plasmas"</i> , Niels Bohr Institute, Copenhagen – Denmark.
Apr 2015	■	School <i>"Scientific Writing for Young Astronomers"</i> , EDP Sciences, Puerto Varas – Chile.
Jun 2014	■	School: <i>"Galaxies: Near and Far, Young and Old"</i> , Vatican Observatory – Vatican City.
Jan - Mar 2013	■	Research Intern , Cerro Tololo Inter-American Observatory - La Serena, Chile. Adviser: Dr. Percy Gómez.

Invited Conference Talks

May 2025	■	19th International Max Planck Research School Symposium - Germany.
Jul 2024	■	17th Marcel Grossmann Meeting - Pescara, Italy.
Feb 2024	■	Quantum Universe Postdoctoral Grant Prize talk at DESY - Hamburg, Germany.
Apr 2023	■	Galactic Center Workshop - Granada, Spain.

Conference Oral Contributions (since 2021)

Sep 2026	■	Annual Meeting of the Astronomical German Society - Garching, Germany.
Aug 2026	■	<i>"Tidal Disruption Events and Nuclear Transients Vol. 2"</i> Conference - Heraklion, Greece.
	■	<i>"Extreme Accretion Events in Supermassive Black Holes"</i> and <i>"Understanding Black Hole Physics through Multi-scale Accretion and Ejection Properties"</i> Scientific Events at 46th COSPAR Assembly - Florence, Italy.
May 2026	■	IAU Symposium 405: <i>"Traversing the Galactic Center in Space and Time"</i> - Brno, Czechia.
Nov 2025	■	RAGTIME27 Workshop - Opava, Czechia.
Jun 2025	■	<i>"X-ray QPEs and Repeating Nuclear Transients"</i> Workshop - Madrid, Spain.
Sep 2024	■	<i>"X-ray transients, theory and observational perspectives"</i> Conference - Warsaw, Poland.

Conference Oral Contributions (since 2021) (continued)

- *"Tidal Disruption Events and Nuclear Transients"* Conference - Heraklion, Greece.
- Apr 2024 ■ RAMSES User Meeting 2024 - New York, USA.
- Dec 2023 ■ *"Fundamental Physics at the Galactic Centre"* Workshop - Porto, Portugal.
- Jul 2023 ■ *"Black-hole accretion discs 50 years on: current fortunes and new challenges"* and *"Supermassive Black Hole Transients and Their Host Galaxies"* Special Sessions at EAS - Kraków, Poland.
- Apr 2023 ■ *"Quantum Universe Day"* at DESY - Hamburg, Germany.
- Nov 2022 ■ 6th Users' Conference of IT4Innovations - Ostrava, Czechia.
- Oct 2022 ■ RAGTIME24 Workshop - Opava, Czechia.
- Sep 2022 ■ 31st Texas Symposium on Relativistic Astrophysics - Prague, Czechia
- Nov 2021 ■ *"The Predictive Power of Computational Astrophysics as a Discovery Tool"* Symposium - Online.
- Sep 2021 ■ RAGTIME23 Workshop - Opava, Czechia.
- Feb 2021 ■ 43rd COSPAR Assembly - Online Symposium.
- Jan 2021 ■ XVI Chilean Astronomical Society (SOCHIAS) Annual Meeting - Online.

Invited Talks at Research Institutions (since 2021)

- May 2026 ■ Special seminar at Institute of Theoretical Physics, Charles University - Prague, Czechia
- Apr 2026 ■ Observatory Colloquium at Hamburg Observatory - Hamburg, Germany.
- Mar 2026 ■ Astronomy seminar at Universidad Federico Santa María - Valparaíso, Chile.
- Aug 2025 ■ Seminar at Astronomical Institute, Czech Academy of Sciences - Prague, Czechia.
- Mar 2025 ■ Astronomy seminar at Universidad de Santiago de Chile - Santiago, Chile.
- Astronomy seminar at Universidad de Concepción - Concepción, Chile.
- Feb 2025 ■ Institute seminar at Max Planck Institute for Astrophysics - Garching, Germany.
- Jan 2025 ■ Seminar at Departamento de Astronomía, Universidad de Chile - Santiago, Chile.
- Apr 2024 ■ Astrophysics seminar at University of Delaware - Newark, USA.
- Jan 2024 ■ Astrophysics seminar at Ludwig Maximilian University - Munich, Germany.
- Oct 2023 ■ Astronomy seminar at Universidad de Valparaíso - Valparaíso, Chile.
- Astronomy seminar at Universidad Federico Santa María - Valparaíso, Chile.
- Sep 2023 ■ Astronomy colloquium at Universidad de Concepción - Concepción, Chile.
- Jul 2023 ■ Astrophysics seminar at Ludwig Maximilian University - Munich, Germany.
- Jun 2023 ■ Observatory Colloquium at Hamburg Observatory - Hamburg, Germany.
- Oct 2021 ■ Relativistic Seminar at Silesian University - Opava, Czechia.
- Sep 2021 ■ Astronomy seminar at Masaryk University - Brno, Czechia.
- Mar 2021 ■ Stellar seminar at Czech Academy of Sciences - Online.
- Feb 2021 ■ Relativistic seminar at Charles University - Online.

Outreach Efforts

- May 2025 ■ *"Lunch with an Astrophysicist"* at Max Planck Institute for Astrophysics, Germany.
- Jul 2024 ■ Galactic Centre Virtual Reality at *"Sternstunden Festival"*, Hamburg Observatory.
- Jun 2024 ■ Galactic Centre Virtual Reality at *"Quantum Universe Day"* at DESY in Hamburg.
- 2023-2024 ■ Editor of the Hamburg Observatory Newsletter.
- Jul 2023 ■ *"Ask an Astronomer"* and Tour Guide at *"Sternstunden Festival"*, Hamburg Observatory.
- 2020 ■ Interview at online newspaper *"El Contraste"* during Solar Eclipse 2020, Chile.

Outreach Efforts (continued)

- 2014-2016  Quantum Physics course for high-school teachers, Pontificia Universidad Católica de Chile.
- 2015  Special lectures for students during the “*Day of Astronomy*” in Curicó, Chile.
- 2013-2014  Physics courses for high-school students at Pontificia Universidad Católica de Chile.
- 2011-2012  Guide of public astronomical observations at Universidad de Concepción, Chile.

References

Prof. Dr. Andreas Burkert

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